SAMPLE

Earth's Sky: Eclipses

When Shadows Fall on Worlds

Aira Astralis

Free sample: Introduction & Chapter 1

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Introduction

Twice a year, on average, somewhere on Earth the Sun goes out in the middle of the day. The Moon slides across it, the sky deepens to twilight, the planets and bright stars appear, the horizon glows orange all around, and for a few minutes the Sun stands in the sky as a black circle crowned by pale streamers that no daylight ever shows. Then the light returns as suddenly as it left, and the world picks up where it stopped. The other half of the story happens at night, more often and less dramatically: the full Moon slides into the Earth's shadow and dims to a dull ember for an hour or two. These are eclipses, and they are the oldest recurring astronomical event that unaided eyes can watch, feared for most of history and understood, at last, well enough to be predicted to the second.

This book is about eclipses as the Earth's sky presents them: what they are, how they were noticed and tamed, what they taught, and what they still teach. The subject sits oddly in modern astronomy. On one hand, an eclipse is pure geometry, arrangement, a consequence of circles and tilts that a schoolroom can demonstrate with a lamp and two balls; nothing about it is mysterious. On the other hand, the few minutes of a total solar eclipse remain, four centuries into the telescope era, the only time the Sun's outer atmosphere can be seen against a dark sky by anyone standing on the ground, and each one still empties cities into the countryside and fills aircraft with spectrographs.

Eclipses also carry a history no other sky event matches. The Babylonians turned them into arithmetic; the Chinese bureaucratized them into dynastic record-keeping; a recorded eclipse may have stopped a war; one supplied the first confirmed element found off the Earth; one settled, in 1919, whether light bends as Einstein said. The Moon's shadow is a periodic visitor with an appointment book centuries long, and every civilization that kept a calendar eventually learned to read it.

The chapters ahead follow that long arc. They begin with the ancient records and the slow discovery that the frightening omen obeys a rhythm, follow the geometry once the heavens were measured, and then turn to the nineteenth-century decades when eclipses briefly became the hottest subjects in science, when every total eclipse emptied observatories into expeditions. They end in the present, when the shadow's exact path can be posted years in advance, when the corona has been touched by spacecraft, and when millions of people plan their summers around two minutes of darkness, because those two minutes, everyone now knows, are worth the trip.

Chapter 1: The Discovery of Eclipses

The discovery of eclipses is not a single moment but a three-stage process: noticing that they recur, recording when they recur, and finding the pattern in the records. Each stage took centuries.

The noticing was ancient and easy. A total solar eclipse is impossible to miss; even a deep partial one dims the light enough to alarm anyone outdoors, and a lunar eclipse turns the full Moon, humanity's fixed nocturnal landmark, the color of rust. The records begin almost with writing itself. Chinese court astronomers were charged to announce eclipses before they happened, and their failures entered legend: an old tradition holds that two astronomers, Hsi and Ho, drank through their duties and were executed when an eclipse surprised the court, a story that at minimum proves that eclipse prediction was a formal obligation of the office by the time the legend was written down. Babylonian scribes, more prosaically, logged everything; the clay tablets of the late Babylonian period run to hundreds of eclipse records, observed and predicted, with times, magnitudes, and the sky's behavior noted in the flat voice of the ledger.

The Chinese corpus deserves one more note, because it contains the field's oldest artifacts. The oracle bones of the Shang dynasty, incised questions put to ancestors and dated by reign, include eclipse records from the twelfth century BC and earlier, which makes them among the oldest datable astronomical observations in existence; the later dynastic histories, compiled with official mandate, kept the sequence going, so that China holds the world's longest continuous eclipse archive. The Islamic astronomical tradition added its own institutional layer: the eclipse prayer institutionalized a calm response, and the observatories of the medieval Islamic world, Maragha and Samarkand among them, refined the lunar tables that made prediction a working tool from Spain to Central Asia, a chain of custody the European tables later inherited through translation.

The Babylonian contribution was the decisive one, because they did not merely record; they computed. Their scribes had noticed that eclipses of the Moon come in families: if one occurs, another will follow 18 years and about 11 days later, at a similar place in the sky. The cycle, three eclipse seasons of 223 lunar months, they built into their predictive arithmetic; a later Greek borrowing of the same period gave it the name by which it is still

called, the saros. The Babylonians' eclipse tables, drawn from records rather than from any theory of orbits, could warn that an eclipse was possible and even roughly when, which was prediction enough to shatter the idea that the darkness was a whim.

The famous stopping of a war belongs to this same phase of the story.

Herodotus reports that day turned to night while the armies of Lydia and Media were locked in battle, and that the two sides, taking this as a divine verdict, laid down their weapons and made peace; the battle is traditionally placed in 585 BC, and the eclipse is one modern calculation identifies with the total eclipse of May 28 of that year, though the identification rests on trusting Herodotus's dates, and scholars still argue about it. What matters for the history is the story's second act: the tradition that Thales of Miletus, the first of the Greek natural philosophers, had foretold the eclipse.

The report is almost certainly wrong in any strict sense, since predicting a solar eclipse at a given place requires more geometry than sixth-century Miletus had, but it may preserve a real achievement: Thales, in contact with Babylonian learning, plausibly knew the saros arithmetic and could have said that a year in that window was eclipse-possible. The legend, in any case, set the pattern for how the Greek world saw the event: the sky behaving lawfully, and the philosopher who knows the law outranking the king who does not.

The Greeks then supplied the third stage, the explanation, and with it a story that runs through this whole book: the moment an eclipse becomes geometry, it stops being a sign. Once the Earth's round shadow, curved on the Moon's face, had shown the Earth's shape, and once the arrangement of the two luminaries and their nodes was worked out, the pattern of eclipses was not a mystery but a clock. By the time of Hipparchus and Ptolemy, eclipse records were being used for a subtler purpose, measuring the Moon's motion, and the great ancient observer Hipparchus built his lunar theory partly by triangulating from old Babylonian timings. From that point forward, the history of eclipse discovery is really two intertwined histories: the refinement of the clock, and the extraction of science from the few minutes when the shadow lands on prepared ground.

A final discovery closes the arc, and it is one of the strangest in the sky: the total solar eclipse, the most terrifying event in the old catalog, is a coincidence of sizes. The Sun is about 400 times wider than the Moon and almost exactly 400 times farther away, so their disks match in the sky to within a few percent. If the Moon were smaller, or farther, there would be only partial and annular eclipses; the crown of streamers would never appear to anyone standing on Earth. Nobody discovered this until the measurements were good enough to see it, and when they did, the coincidence acquired a melancholy footnote that the final chapters return to: the Moon is drifting slowly away, and the perfect match is a phase in a long story, not a permanent arrangement.

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